

ABDELRAHMAN ELMOGY

Data Analyst | Automation Engineer

Cairo, Egypt | +20 101 427 7880 | Abdelrahmannelmogy22@gmail.com | LinkedIn: Abdelrahman Elmoghy

Portfolio: abdelrahmannelmogy.github.io

PROFESSIONAL SUMMARY

Data Analyst and Automation Engineer with **3+ years of experience** turning complex datasets into clear, actionable insights using **Power BI, SQL Server, Python, and Excel**. Strong track record of **designing dynamic Power BI dashboards** for real-time KPI monitoring, **building Python automation pipelines** that cut manual effort by 60%, and delivering end-to-end reporting, campaign analytics, and customer behavior analysis. Also proven in **designing and scaling AI voice agent systems** (28 assistants, 85,000+ daily calls) and **migrating cloud infrastructure to on-premise servers**. Combines analytical depth with engineering capability to deliver measurable, data-driven results.

CORE COMPETENCIES

✓ Power BI Dashboard Design	✓ Python Data Analysis	✓ SQL Server & Data Modeling
✓ KPI Tracking & Monitoring	✓ Python Automation & ETL	✓ Excel (Advanced)
✓ AI Voice Agent Systems	✓ Flask/FastAPI API Development	✓ Campaign Analytics
✓ Data Visualization	✓ Server & Infrastructure Mgmt	✓ Process Optimization

PROFESSIONAL EXPERIENCE

Data Analyst & Automation Engineer

Maharah International for Debt Collection — Cairo, Egypt *July 2024 – Present*

Data Analysis & Reporting

- **Designed and delivered interactive Power BI dashboards** (Debt Recovery Performance Analytics, Campaign ROI Analysis, Call Center Operations Intelligence, Collection Recovery by Customer Classification) used daily by management and operations teams, replacing static Excel reports with real-time visual analytics that directly shaped collection strategies and resource allocation.
- **Built Python data pipelines** for automated data extraction, transformation, and loading (ETL) from SQL Server, reducing reporting cycle from days to hours.
- **Analyzed debt recovery trends and customer behavior patterns** using Python (Pandas, NumPy) and Power BI, providing insights that directly boosted collection recovery rates.
- **Automated reporting pipelines** using Python and SQL Server, cutting manual effort by 60% and significantly improving data accuracy and consistency.

AI Voice Agent Development & Scaling

- **Designed, built, and scaled an AI Voice Agent system** comprising 28 intelligent voice assistants handling 85,000+ daily outbound calls for major telecom clients (Zain, STC, Mobily), generating over 1,000,000 minutes monthly.
- **Built custom APIs (Flask/FastAPI)** connecting AI voice agents to SQL Server databases for real-time customer data retrieval, phone normalization, and live call context injection during active conversations.
- **Managed enterprise-level vendor negotiations** with Vapi AI for scaling voice operations, evaluating concurrency, cost optimization, and enterprise agreements for 400,000+ minute monthly commitments.

Campaign Analytics & Automation

- **Led end-to-end performance analysis** of IVR, WhatsApp, and email campaigns using Power BI, SQL Server, and Python — delivering data-driven optimization strategies.
- **Developed internal tools** (Claim Document Search Engine, Identity Retrieval System, Automated Payment Verifier, Campaign Automation Pipeline) built with Python Flask, all connected to company databases for live data access.
- **Integrated WhatsApp Business API** (Evolution API + Chatwoot) on VPS infrastructure for multi-agent customer communication and debt collection follow-ups.

Technical Leadership & Infrastructure

- **Provided technical leadership** for the electronic collection team, guiding automation initiatives and system improvements to raise operational efficiency.
- **Migrated critical systems from cloud-hosted (cPanel) environments to on-premise Windows Server infrastructure**, deploying Flask/FastAPI applications locally, reducing latency and cutting hosting costs.

System Administrator & Data Analyst

Royal Canadian School — New Administrative Capital, Egypt *Aug 2023 – June 2024*

- **Analyzed and interpreted complex datasets** to provide actionable insights supporting school management decisions.
- **Built data management systems** using Excel and SQL, ensuring accuracy and reliability of institutional data.
- **Created budget tracking and consumption reports** using Excel, enabling accurate financial forecasting.
- **Collaborated with staff** to identify IT and data needs, delivering tailored solutions that improved operational workflows.

Freelance Python Developer

Remote *Oct 2023 – Present*

- **Built custom Python automation scripts** for clients across multiple industries, reducing manual workloads by up to 40% and saving hundreds of working hours monthly.
- **Designed and deployed web scraping systems** (BeautifulSoup, Selenium) to extract, clean, and structure data from complex websites into ready-to-use datasets and reports.
- **Delivered end-to-end data analysis solutions** using Python (Pandas, NumPy) and Power BI, turning raw client data into visual dashboards and actionable business insights.
- **Developed standalone Python tools** (Excel Automation Toolkit, Database Integration Bridge, PDF Batch Processor) that automate repetitive workflows and are still in active use by clients.

KEY PROJECTS

AI-Powered Voice Agent Platform

- **Designed and deployed 28 AI voice assistants** for automated debt collection across Saudi Arabia, handling 85,000+ daily outbound calls and generating 1M+ minutes/month for Zain, STC, and Mobily.
- **Built real-time customer lookup APIs (Flask/FastAPI)** integrated with SQL Server for live call data retrieval, phone normalization, and context injection during active conversations.
- **Negotiated enterprise agreements with Vapi AI** for 400,000+ minute monthly commitments, optimizing concurrency and cost at scale.
- **Resolved Arabic TTS quality issues**, API latency bottlenecks, hallucination control, and concurrency scaling challenges in production.

Campaign Development & Analytics

- **Led end-to-end development and analysis** of IVR, WhatsApp, Email, and Voice Agent campaigns — fully connected to company databases and call center systems.
- **Built Campaign Automation Pipeline** — Python-based automation tool for scheduling, executing, and tracking multi-channel collection campaigns.
- **Delivered data-driven optimization strategies** using Power BI, SQL Server, and Python, directly improving campaign engagement and recovery rates.

Infrastructure & Server Deployments

- **Migrated critical applications** from cloud-hosted cPanel to on-premise Windows Server, deploying Flask/FastAPI services locally.
- **Deployed Claim Document Search Engine** — Flask-based document search system for 100M+ claim files on local network.
- **Deployed Identity Retrieval System** — identity-based contract and invoice document retrieval system.
- **Set up WhatsApp Integration** (Evolution API + Chatwoot) on VPS with PostgreSQL & Redis for multi-agent communication.

Power BI Dashboards

- Debt Recovery Performance Analytics
- Multi-Channel Campaign ROI Analysis
- Institutional Budget & Expenditure Analytics
- Call Center Operations Intelligence

SQL Server

- IVR Engagement & Conversion Analytics
- Payment Trend & Lifecycle Analysis
- Client Data Normalization Engine
- Customer Risk Profiling & Segmentation

Python Automation & Analysis

- Digital Fraud Detection & Pattern Analysis
- Collector Commission Calculation Engine
- Campaign Automation Pipeline
- Automated Payment Verification System

Excel & Reporting

- Annual Resource Consumption Forecast
- Workforce Attrition & Retention Analysis
- Segment-Based Collection Impact Study
- Data Quality & Update Impact Assessment

Python Tools

- Excel Automation Toolkit
- Database Integration Bridge
- Data Type Conversion Engine
- PDF Batch Processing Tool

Python Projects

- Automated Data Sync & Update System
- Startup Growth & Scaling Analysis
- Appointment Scheduling Optimization
- Claim Document Search Engine

TECHNICAL SKILLS

Data Analysis & Reporting: Python (Pandas, NumPy, Matplotlib), SQL Server, Power BI, Excel (Advanced)

Automation & Scripting: Python Scripting, Web Scraping (BeautifulSoup, Selenium), Workflow Automation, API Integration

AI Voice Agent Engineering: Vapi AI Platform, Voice Agent Design & Scaling, SIP/PBX Integration (Issabel/Asterisk), IVR Systems

Database Management: SQL Server Administration, MySQL, Data Modeling, Query Optimization

Visualization & BI Tools: Power BI (DAX, Power Query, Data Modeling), Excel Pivot Tables & Charts

Infrastructure & Tools: Git, cPanel, Windows Server, Linux (Ubuntu), Flask/FastAPI, VPS Management, On-Premise Deployment

Soft Skills: Problem-Solving, Team Leadership, Technical Guidance, Time Management, Attention to Detail

CERTIFICATIONS

Feb – Apr 2023 Udacity — Data Analysis Professional

Nov – Dec 2023 Planet of Skills — Excel + Power BI Diploma

Mar – Apr 2022 Udacity — Data Analysis Challenger

Sep – Nov 2022 HackerRank — Advanced & Intermediate (Python, SQL)

Aug – Sep 2022 NTI — Cyber Ops Associate

Jul – Sep 2021 ITI — CCNA

EDUCATION

Bachelor of Communication Engineering

Mansoura University, Egypt | 2018 – 2023

LANGUAGES

Arabic — Native **English** — Intermediate **German** — Beginner